**Other reflexes**

Reflexes present at the time of birth and remain active throughout life time. For example, the pupil reflex, sneeze reflex, blinking reflex, cough reflex and yawn reflex.

### Twelve Cranial Nerves

Remember the mnemonic: **O**ne **O**ld **O**wl **T**urned **T**op, **A** **F**at **A**unt **G**o **V**iewed **S**ome **H**op

- **Olfactory**: smell
- **Optic**: vision
- **Oculomotor**: eye movements.
- **Trochlear**: eye movements.
- **Trigeminal**: chewing and sensory to face
- **Abducens**: eye abduction.
- **Facial**: facial expressions, the tongue and the salivary glands.
- **Acoustic/Auditory**: hearing
- **Glossopharyngeal**: taste impulses and the secretion of saliva from the parotid gland.
- **Vagus**: respiratory and the digestive systems such as the pharynx and swallowing.
- **Spinal accessory**: movement of shoulder.
- **Hypoglossal:** Tongue movement.

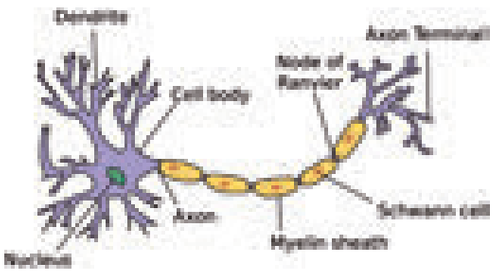
### Functional Units of Nervous System

The nervous system is comprised of millions of neurons and glial cells.

### Types of nerves

- Sensory or afferent nerves
- Motor or efferent nerves
- Mixed nerves

## Neuron




**Neuron**

The neuron is the primary type of nerve cell in the nervous system. Glial cells forms myelin sheath over the neuron for protection.

Neurons communicate with other neurons by sending impulses at the synapse, a point of junction between any two neurons. There are two types of neurons which are sensory neurons and motor neurons. Sensory neurons sense and transmit information such as taste, touch and sight. Motor neurons send and transmit messages that involve muscle movement.

### Glial Cells

Glial cells are non-neuronal cell that provide support and nutrition, maintain homeostasis, form myelin and patricipate in signal transmission in the nervous system.

### Types of nerves

**Sensory or afferent nerves**When action potentials are generated by sensory receptors on the dendrites of these neurons, they are transmitted to the spinal cord by the sensory nerve fibres. The impulses may then pass to the brain or to connector neurons of reflex arcs in the spinal cord motor or efferent nerves.

**Motor nerves** originate in the brain, spinal cord and autonomic ganglia. They transmit impulses to the effectors organs muscles and gland. There are two types:

**Somatic Nerves** – involved in voluntary and reflex skeletal muscle contraction.

**Autonomic Nerves** (sympathetic and parasympathetic) – involved in cardiac and smooth muscle contraction and glandular secretion.

**Mixed Nerves:** In the spinal cord, sensory and motor nerves are arranged in separate groups or tracks. Outside the spinal cord, when sensory and motor nerves are enclosed with the same sheath of connective tissue they are called mixed nerves.



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